

Amir Reza Khatibi

Data Science & Engineering Student

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Profile

Student in **Data Science and Engineering** with a strong background in **machine learning**, **statistical modeling** and **applied data analysis**. Experienced in building end-to-end ML pipelines and transforming quantitative insights into interpretable and business-oriented solutions.

Education

- 10/2025 – 10/2027 **Master's Degree in Data Science and Engineering**, *Politecnico di Torino*, Turin, Italy
Focus on machine learning, data engineering, statistical modeling and large-scale data analysis.
- 10/2022 – 10/2025 **Bachelor's Degree in Computer Engineering**, *Politecnico di Torino*, Turin, Italy
- Final grade: **110/110**
 - Selected member: **INTRAPRENDENTI Project** (Honors School)
 - Member: **IEEE-Eta Kappa Nu** (Mu Nu Chapter)
 - Relevant coursework: Algorithms and Data Structures, Signals and Systems, Databases, Operating Systems, Computer Networks, Probability and Statistics, Linear Algebra

Experience

- 10/2025 – 01/2026 **Teaching Assistant – Signals and Systems**, *Politecnico di Torino*, Turin, Italy
- Supported more than **300** students through office hours and laboratory sessions
 - Explained the mathematical foundations of signal processing and linear systems
- 10/2024 – 12/2024 **Laboratory Teaching Assistant – Algorithms and Data Structures**, *Politecnico di Torino*, Turin, Italy
- Assisted more than **200** students in implementing fundamental algorithms
 - Analysis of algorithmic complexity, recursion, sorting algorithms and problem solving
 - Code debugging and structured programming best practices

Projects

Real Estate Price Prediction (GitHub link)

- End-to-end regression pipeline including data cleaning, EDA and feature engineering
- Model comparison (Linear Regression, HistGradientBoosting, Random Forest) with cross-validation
- Evaluation using RMSE, MAE and adjusted R^2
- Implementation using Pandas, NumPy, Matplotlib and Scikit-learn

News Article Topic Classification (GitHub link)

- Developed a multi-class news article classification system (7 categories) using NLP techniques
- Built a preprocessing pipeline with TF-IDF representation based on word and character n-grams (~65k features) and integration of metadata (cyclic timestamp encoding and page rank)
- Implemented a Linear Support Vector Classifier (LinearSVC) with class weights to address class imbalance
- Compared models (Logistic Regression, linear ensembles, SVD + XGBoost) and performed hyperparameter tuning

Skills

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|-------------------|---|
| Programming | Python, C, C++, Java |
| Data Science & ML | NumPy, Pandas, Matplotlib, Scikit-learn, PyTorch |
| Databases | SQL, MongoDB; Data Warehouse design |
| Tools | Git, Linux, Jupyter Notebook, MATLAB, Microsoft Office, PowerBI, Docker, AWS, FastAPI |